

	Survived	Embarked	Age	AgeNA	SibSp	Parch	Sex	Fare	Survived (one-hot)
7	0	1	54	0	0	0	male	51.8625	S
8	0	3	2	3	1	0	male	21.075	S
9	1	3	27	0	2	0	female	11.1333	S
10	1	3	4	1	1	0	female	16.7	S
11	0	3	20	0	0	0	male	8.05	S
12	0	3	39	1	5	0	male	31.275	S
13	0	3	14	0	0	0	female	7.8542	S
14	1	2	55	0	0	0	female	16	S
15									

↓

602 rows

8. Assemble feature columns into a feature vector (for use with the pipeline)

- For the `Age` column, include both `AgeFilled` and `AgeNA`.
- For `Sex` and `Embarked`, include only the encoded versions produced by `OneHotEncoder`.

29

```
# # Assemble all feature columns into a single 'features' vector
from pyspark.ml.feature import VectorAssembler

va = VectorAssembler(
    inputCols=[
        "Pclass",
        "AgeFilled",
        "AgeNA",
        "SibSp",
        "Parch",
        "Fare",
        "Sex_ohe",
        "Embarked_ohe"
```